GLASS-BOX GREEN INSIGHTS

Lloyds Megathon Problem Statement

A simple guide to what we need to build, how it works, and how five people can divide the work.

ML predicts. Rules decide. The carbon engine calculates. The LLM explains.

Main goal

Find customers who may benefit from green financial products, estimate the possible sustainability impact, and explain every recommendation clearly and safely.

This guide focuses first on the four required products: EV Finance, Green Home Loan, Solar Finance, and Paperless Banking.

1. The problem in simple words

Banks have lots of useful information in transactions, product records, energy usage and digital behaviour. The challenge is to turn that information into useful green opportunities without creating a black-box system.

Our platform should answer five simple questions for every recommendation:

- Which green product could help this customer?
- Why was this customer selected?
- Is the product allowed to be shown to them?
- What could the environmental benefit be?
- What should the customer or relationship manager do next?

We are not building only a chatbot or only an ML model. We are building a complete, traceable decision pipeline.

Required output

For every customer-product pair, return a propensity score, a separate confidence score, fixed reason codes, eligibility status, an impact estimate when enough data exists, and a plain-language explanation.

MOST IMPORTANT IDEA

2. The golden rule

Part	Job	Must never do
ML models	Predict product take-up and categorize transactions.	Decide eligibility or calculate carbon.
Eligibility rules	Decide whether a product may be shown.	Be overridden by a high ML score.
Carbon engine	Calculate impact from formulas and documented factors.	Ask an LLM to invent a value.
LLM	Explain approved structured results in simple language.	Predict, calculate, or decide eligibility.

Example

The ML model says there is an 84% chance that a customer may take EV Finance. The rules confirm that the offer may be shown. The carbon engine estimates a possible saving of 900–1,200 kg CO_2e per year. Only then does the LLM turn those facts into a friendly explanation.

A high propensity does not mean the customer is eligible.

BUILD THESE FIRST

3. The four required opportunities

Product	Useful signals	Impact / outcome
EV Finance	High fuel spend, high mileage, existing petrol/diesel vehicle or vehicle finance.	Compare current fuel emissions with estimated EV electricity emissions.
Green Home Loan	Homeowner or mortgage customer, high utility use, home-improvement activity.	Estimate improvement from an agreed retrofit such as insulation or efficient heating.
Solar Finance	Home ownership, high electricity use, no existing solar system.	Estimate grid electricity displaced by solar generation.
Paperless Banking	Paper statements plus regular mobile/web banking activity.	Report statements/pages moved to digital separately from CO2e.

Important product boundaries

- Define exactly what Green Home covers so it does not clash with Solar Finance.
- Do not recommend EV Finance to an existing EV owner unless a valid replacement case exists.
- Do not call digital adoption a carbon saving unless a documented method supports it.
- Optional Green Investments and Green Loans come only after the four required flows work.

4. Data and Customer 360

Real customer banking data must not be used. Create realistic synthetic customers and connect all tables with customer_id.

Table	Example information
Customers	Age band, income band, postcode, home ownership, preferences
Transactions	Date, merchant description, amount, debit/credit
Products	Current accounts, mortgage, loans, investments
Vehicles	Fuel type, vehicle age, annual mileage
Energy usage	Monthly electricity and gas usage
Digital usage	Paper statements, app use, digital payments, branch visits
Reference data	Green product rules and versioned emissions factors

Make the data believable

- Create customer types such as a high-mileage commuter, suburban homeowner, solar-ready homeowner, and digitally active paper customer.
- Use hidden factors such as green interest and financial readiness, plus noise, to generate future product take-up.
- Choose a cutoff date. Features must be from before it; take-up labels should come from a later window, such as 90 days.
- Split train, validation and test data by customer to prevent leakage.

5. Models and recommendations

First categorize every transaction into exactly one of eight groups: Fuel, Electricity, Gas, Public Transport, Home Improvement, Automotive, Utilities or Other.

Recommended transaction approach

- Use clear merchant rules for easy cases such as petrol stations, rail operators and energy suppliers.
- Use TF-IDF plus Logistic Regression for unclear merchant descriptions.
- Document difficult boundaries, such as Fuel versus Automotive and energy suppliers versus general Utilities.

Recommended opportunity approach

- Train one model for each required product. LightGBM or XGBoost is suitable; Logistic Regression is a good baseline.
- Calibrate probabilities so that a score such as 0.80 behaves like a meaningful probability.
- Map important model features to fixed reason codes such as HIGH_FUEL_SPEND or PAPER_STATEMENTS_ACTIVE.

Keep these separate

Propensity	Chance of product take-up
Confidence	How reliable the prediction is, based on data quality and model certainty
Eligibility	Whether the product rules allow it to be recommended

NO HIDDEN DECISIONS

6. Rules and sustainability calculations

The eligibility engine runs before a product reaches the customer view. Every pass, failure or missing-data result should have a code and rule version.

Example EV calculation

Current emissions = fuel used per year × fuel factor
EV emissions = EV electricity per year × grid factor
Potential reduction = current emissions – EV emissions

Every impact output should include

- A lower and upper estimate instead of false precision.
- The unit and time period, for example kg CO2e per year.
- Formula or method name.
- Assumptions, such as annual mileage and vehicle efficiency.
- Emission-factor source, year and version.
- A clear 'insufficient data' result when inputs are missing.

Keep CO2e, energy, paper and digital-adoption results separate. The same input must always produce the same output, so these calculations should be normal code with unit tests—not an LLM prompt.

7. Safe customer explanations

The LLM receives only an approved JSON payload containing the product, eligibility result, scores, reason codes, impact range, action and timestamp.

After writing the message, a verifier checks:

- Every number exists in the input payload.
- The product name is correct.
- No new reason or unsupported promise was added.
- The message uses words such as estimated or potential.
- Missing information is stated honestly.
- An ineligible product is never presented as an offer.

If verification fails, show a safe fixed template instead of the LLM answer.

Example message

“Your recurring fuel spending and annual mileage suggest that an electric vehicle may suit your travel pattern. The estimated potential reduction is 900–1,200 kg CO2e per year, based on documented fuel and electricity factors. Explore EV options, subject to eligibility checks.”

8. What the app should show

Customer screen

- Recommended product and one clear action.
- Why it is relevant in plain language.
- Propensity, confidence and evidence.
- Estimated impact range, method and assumptions.
- Data timestamp and eligibility note.

Business dashboard

- Relationship-manager queue: who to contact and about which product.
- Opportunities by product, region and customer segment.
- Potential CO2e and energy impact shown as ranges.
- Paper and digital-adoption opportunities shown separately.
- A drill-down path: Segment → Customer → Opportunity → Evidence → Impact → Action.
- An audit view showing why a product was blocked.

Suggested priority score

Propensity × confidence × normalized potential impact

Always show the individual values as well, so the combined score does not become another black box.

CLEAR OWNERSHIP

9. Five-person work division

Person 1 — Data & Customer 360

Generate synthetic tables, labels, cutoff dates, customer-level splits, data dictionary and the final feature table.

Person 2 — ML & Recommendations

Build the transaction classifier, four propensity models, baselines, calibration, confidence method, reason codes and model metrics.

Person 3 — Rules & Carbon Engine

Build product catalogue, eligibility rules, emissions-factor table, impact formulas, ranges, abstention logic and unit tests.

Person 4 — LLM & Responsible AI

Build the structured prompt, verifier, template fallback, explanation evaluation, reason-code wording and Responsible AI note.

Person 5 — App & Integration

Build FastAPI/Streamlit, integrate all components, create customer/business/audit views, architecture diagram, README and demo fallback.

All five people must agree on one shared input/output JSON format before building.

10. Build plan

Phase	What to complete
1. Agree	Four products, schemas, cutoff date, outcome window, product rules, reason codes and demo customers.
2. One full flow	Make EV Finance work from raw data to verified dashboard explanation.
3. Expand	Reuse the same pattern for Green Home, Solar and Paperless.
4. Test	Baselines, held-out metrics, leakage checks, missing data, conflicts, ruled-out products and LLM failures.
5. Polish	Dashboard styling, regional views, reports, architecture, three-minute demo and cached fallback.

If time runs short

Cut optional products, customer memory and extra visual polish first. Never cut the eligibility gate, documented impact calculations, baseline comparisons or the LLM verifier.

Shared result format

Each opportunity should contain: customer ID, product, propensity, confidence, reason codes, eligibility result, decision codes, impact status/range/unit/method, data timestamp, model version, rule version and factor version.

11. Testing and scoring

Part	What to report
Transaction classifier	Per-class F1, macro-F1, confusion matrix and keyword-rule baseline.
Propensity models	PR-AUC, lift, calibration, precision at top customers and a simple baseline.
Carbon engine	Hand-calculated examples, unit tests, units, factors and missing-input tests.
LLM	Grounded-output rate, correct product, unsupported-number rate, completeness and fallback rate.
Responsible AI	Privacy, bias checks, explainability, hallucination controls, human oversight and data limits.

Must-have test scenarios

- High propensity but failed eligibility.
- Existing EV owner with no replacement case.
- Missing mileage or energy information.
- Conflicting customer signals.
- Unknown merchant description.
- LLM tries to add a number not in the payload.

Judges may rerun the code, so use fixed random seeds, pinned dependencies and one clear run command.

12. Strong demo ideas

Why not recommended?

Show a customer with high EV propensity whose offer is blocked by the existing-EV rule.

What-if calculator

Let the user change mileage or solar assumptions and see the impact range update. Label it as a scenario.

Evidence panel

Show data date, model version, rule version, formula and emissions-factor source.

Safety console

Run missing-data, conflicting-signal and LLM-hallucination tests live.

Manager queue

Show who should be contacted, about what, why now, possible impact and the next action.

Three-minute demo

Time	Scene
0:00	Customer recommendation with evidence and impact.
0:30	Blocked EV recommendation proves the rules work.
1:00	Manager queue filtered by region or product.
1:40	Drill down from opportunity to formula and assumptions.
2:10	Run missing-data and LLM-verifier scenarios.
2:40	Show held-out results, baselines and architecture.

THE ONE-LINE PITCH

We find relevant green opportunities, check that they are allowed, calculate the possible impact transparently, and explain every result with evidence.

Build the smallest complete and trustworthy system first. Add innovation only after it works end to end.

Source basis

Prepared from the Lloyds Megathon Problem Statement 1 and the team's Glass-Box Green Insights planning deck. This document simplifies the wording; the official problem statement remains the final authority.